

Executive Summary

We propose a high-speed, limited-stop north-south subway running roughly Inwood (northern Manhattan) to Battery Park (southern tip) in ~20 min end-to-end. A probable alignment is under the Manhattan grid (roughly Broadway/Central Avenue), stopping at major transfer hubs (Dyckman/Inwood, 125th St, Columbus Circle (59th St), 34th St/Penn, 14th/Union Sq, Fulton/World Trade Center area, Battery Park). We estimate ~8 major stations (including termini) spaced 2–4 km apart, connecting to almost all existing lines and major ferries. Stations would each serve on the order of 3,000–7,000 morning peak boardings (based on 2019 station ridership data [7†L99-L107]), drawing riders from large catchment areas. Tunnels would mostly lie 20–50 m below ground through Manhattan schist/gneiss (bedrock ~8 m deep at the Battery up to ~45 m deep around City Hall [28†L612-L620]). Bored tunnel (double-track tubes) using heavy-rail TBMs (e.g. double-shield or EPB machines as used on East Side Access [31†L202-L205]) is appropriate. Ventilation and emergency egress would use station shafts and intermediate shafts/vents, with modern fan systems at surface or deep shafts.

We assume a modern high-acceleration EMU fleet (similar to London's Elizabeth line) capable of ~2 m/s² accel and ~150 km/h max speed, running ~3–4 min headways to achieve 20-min run. Estimated capacity is ~30–50k pphpd. A new depot/yard would likely be built at one end (north or south) or use/expand an existing yard with new access tracks. CBTC signaling and possibly positive train control would be used. Power would likely be standard 625–750 V DC third rail (with heavy power draw, possibly boosted feeder stations) since all NYC subways use it.

Design-build procurement (with heavy use of PPP elements or joint ventures to share risk) is likely; the MTA's recent practice (e.g. SAS Phase 2 [13†L144-L152] [19†L233-L240]) favours large DB contracts. A phased delivery (EIS/Design, early works, TBM tunnelling, station construction, systems fit-out, testing) would span roughly 12–15 years. We estimate roughly ~\$70–80 billion USD (2026\$) total (including ~30–40% contingency) – roughly £50–60 billion at 1 USD ≈ 0.75 GBP. Rough cost breakdown: tunnels (~\$45B), stations (~\$10–15B), systems and integration (~\$5B), rolling stock (~\$1–2B), land/relocation (~\$0.5B), plus 30–40% contingency [19†L233-L240] .

Regulatory review would be extensive: NEPA/SEQR EIS (similar to SAS2) will be required [48†L70-L77] . Community impact is a concern, but most work is underground; station box construction and street-level shafts would be carefully staged. Mitigation (noise, dust, traffic) would follow standard practice (e.g. enclosure, limited hours). We identify major risks (geotechnical surprises, utility conflicts, cost overruns – common on NYC mega-projects [13†L140-L148] [19†L233-L241]) and suggest contingency, early surveys, and insurance (owner-controlled) to mitigate.

Key findings: A Manhattan end-to-end line is technically feasible but extremely challenging and costly. We outline an optimal alignment, stations, tunnels, and rolling stock to meet a 20-min run, drawing on MTA and industry precedent [13†L140-L148] [19†L233-L240] . Environmental/community issues and permitting will be complex (requiring a full federal/state EIS [48†L70-L77]). Our cost estimate (~\$70–80B) and 12–15 yr timeline reflect NYC's high costs and strong regulatory environment. Further study (demand modeling, refined geology) would refine these estimates.

Route Alignment & Station Locations

Our optimal route runs roughly along Broadway/Seventh Avenue (west-central Manhattan) for most of its length. **Northern Terminus:** Inwood–207th St area (1/9 line terminus) – best captures northern Manhattan density and connects to local buses and Columbia University (a short bus). **Southern Terminus:** Battery Park/Whitehall (1 line/4/5 lines) – connects to Staten Island Ferry, downtown ferries, and South Ferry subway (1/4/5) 【7†L100-L107】 .

Between these, major stations (with **approximate cross-streets** and interchanges) might include:

- **207th St/Inwood (N terminus)** – Transfers: 1/9 train, local buses. Ridership: ~2,000 pphpd AM peak boarding (small northern catchment).
- **125th St (Harlem)** – Interchange: 1 (125 St), A/D (125 St), maybe 2/3 (125 St) – serves Harlem. Catchment: high residential/work. Peak AM boardings ~4–6k (ridership surrogates: 125th corridor stations ~30M/yr 【7†L100-L107】).
- **59th St (Columbus Circle)** – Interchange: A, B, C, D, 1 (59th St), multiple bus lines. Catchment: west Midtown/business districts. Peak ~6k. (Columbus Circle sees ~23M/yr 【7†L106-L110】 .)
- **34th St (Herald/Penn)** – Interchange: 1/2/3 (Penn 34), A/C/E (Penn), B/D/F/M (Herald Sq), LIRR/Amtrak (Penn Station), NJ PATH (34th St); also Madison Square Garden and midtown E/W corridors. Catchment extremely high. Peak ~7k. (34th/Penn complex handles ~25–26M/yr on IRT/IND 【7†L104-L110】 .)
- **14th St (Union Square)** – Interchange: 4/5/6 (Union Sq), N/Q/R/W/L (14th St–Union Sq); plus buses/citi-bike. Very high pedestrian volume (32M/yr 【7†L103-L107】). Peak ~7–8k.
- **Chambers/Fulton (Lower Manhattan)** – Interchange: A/C/E (World Trade Center), 2/3 (Fulton St), J/Z (Chambers), PATH (WTC), and ferries (East River). Covers Financial District and WTC. Peak ~6k.
- **Battery Park (South Ferry)** – Interchange: 1, 4/5, Staten Is. Ferry, local buses/ferries. Southern tip catchment (tourists, Staten Island commuters). Peak ~3k.

Table: Candidate Stations and Transfers. (Estimated peak boardings assume Manhattan’s existing busiest station hourly flows 【7†L100-L107】 , scaled for fewer stops on this line):

Station	Location (X-street or coord)	Interchanges (lines, ferries, etc.)	Catchment Area	Est. AM Peak Hour Boardings
Inwood (207 St)	Broadway/ 207th St (40.868N, -73.918W)	NYC Subway 1/9 (DT), buses	Inwood, Washington Heights (north end)	~2,000
125th St	Broadway/ 125th St (40.814N, -73.941W)	Subway 1, A, B, C, D (multiple at 125 St)	Central/West Harlem	~5,000
59th St (CC)	Broadway/59th St (40.768N, -73.982W)	Subway A, B, C, D (59 St– CC); 1; buses	Midtown (Columbus Cir area), UWS, Lincoln Ctr	~6,000

Station	Location (X-street or coord)	Interchanges (lines, ferries, etc.)	Catchment Area	Est. AM Peak Hour Boardings
34th St (Penn)	7th Ave/34th St (40.750N, -73.989W)	Subway 1/2/3 (34), A/C/E (34), B/D/F/M (Herald); LIRR/Amtrak (Penn); PATH	Midtown shopping/offices, universities	~7,000
14th St (Union)	4th Ave/14th St (40.734N, -73.989W)	Subway 4/5/6 (14), L/N/Q/R/W (14)	Union Square area (UGC/QG market); Village	~7,000
Fulton/ Chambers	Broadway/ Fulton St (40.713N, -74.008W)	A/C/E (WTC); 2/3/4/5 (Fulton); J/Z (Chambers); PATH (WTC); ferries (East River)	Financial District, WTC, Civic Center	~6,000
Battery Park	State St/ Whitehall (40.703N, -74.017W)	Subway 1 (South Ferry); 4/5 (Bowling Green); Staten Island Ferry; NY Waterway	Battery Park City, S. Manh.; Staten I ferries	~3,000

Note: Boardings are rough estimates based on extrapolating from 2019 ridership data [7†L100-L107] and assuming the new line draws new trips. Each station's **catchment** covers key jobs/residential areas and connects to existing lines; for example, 34th St Penn (serving Penn Station) saw ~26M entries in 2019 [7†L104-L110] , so even a fraction of those could use the new line in peak hours.

Tunnel Alignment, Depth, and Geology

The tunnel would largely follow Manhattan's length underground, keeping under public right-of-way to minimize property acquisition. Our provisional vertical alignment would start near bedrock (~10 m) at Inwood, descend somewhat (City Hall area has a "bedrock valley" ~45 m deep [28†L612-L620]), then rise under midtown (14th–34th St has relatively shallow rock [28†L612-L620]), and finally descend to ~8–20 m near Battery (bedrock ~8 m deep [28†L612-L620]). The profile is roughly 20–50 m below surface; actual design would follow gravity drainage to avoid water pooling, with gentle grades (~<2%) between stations.

Geology: Manhattan's substratum is mainly Manhattan Schist (metamorphic), with zones of gneiss and occasional pegmatite/granite [31†L202-L205] [30†L169-L176] . Glacial till/sands overlie the bedrock, especially in Midtown below ~14th St, and deeper clay/varved deposits near the Harlem River. Overall, rock quality is good ("hard" 100–275 MPa UCS [31†L202-L205]) but heavily foliated with high horizontal stress [30†L169-L176] . Past projects (East River tunnels, water tunnels, East Side Access) have successfully bored through similar conditions.

Ventilation/Egress: Ventilation would use longitudinal jet fans plus shafts. Each station entrance doubles as an emergency egress and ventilation shaft. Additional mid-tunnel shafts (e.g. near 96th St or 23rd St) would provide pressure relief and smoke extraction in an incident. This is similar to designs on deep systems (see Crossrail/London or SAS stations with large fan plants). Fume extraction and smoke vents in stations would connect to surface vents. Design would follow NFPA 130 and NYC Fire Code; e.g. stations spaced every ~2–3 km exceed the one-exit-per-2 km guideline.

Construction Methods & Staging

Tunneling: Given the long, straight nature of the route, we would use tunnel-boring machines (TBMs) for the main tubes. Likely twin single-bore tunnels (one per direction) of diameter ~7–8 m, precast-segment lined. A double-shield (open/closed mode) hard-rock TBM, or Earth-Pressure-Balance (EPB) machine (for any soft pockets), are candidates [31†L202-L205] [30†L169-L176] . TBMs would be launched from large shafts (e.g. near 207th St and Battery) to bore entire segments. The East Side Access project used Robbins TBMs through Manhattan schist [31†L202-L205] .

Station construction: Stations (8 large underground boxes) will be built via cut-and-cover or box jacking, depending on depth. In deep bedrock sections, stations will be mined/NATM style (as on SAS or Crossrail). Shaft sites for station access (surface boxes) are needed at each major station. Stations may require temporary closures of streets/blocks. Experience from SAS phase 1 shows deep stations with escalator shafts up to 30 m deep.

Shafts & utilities: Seven or more shafts will be needed: two TBM launch/receive shafts (north, south), plus station access shafts. Early in the project, detailed utility mapping is essential. Manhattan has dense utilities (water, sewer, electric, telecom) – many may need relocation or protection. Early “probing” contracts (like SAS2 did [13†L168-L176]) can reduce risk. We assume utility relocation as separate early contracts (e.g. 2028–2030) to move major water, gas, fiber lines from the tunnel path.

Staging: Construction will be staged from both ends and possibly one central location (midtown). Two TBMs could bore from north and south; if a single TBM is used, it might start northbound from Battery and be retrieved/increased at midtown. Muck removal will use conveyor/rail down shafts (like East Side Access [31†L243-L252]). To minimize disruption, station excavation might use sequential cut-and-cover in short sections, and material removed by truck or conveyor. Community mitigation (noise enclosures, off-hours trucking, dust control) will follow the 2nd Av Subway model.

Rolling Stock & Performance

To cover ~21 km end-to-end in 20 min (including stops), trains need high performance. We assume modern electric multiple-units (EMU) with:

- **Acceleration:** ~1.0–1.5 m/s² (2.2–3.4 mph/s). NYC’s R160 subways accelerate ~0.7 m/s²; high-speed EMUs (Crossrail) achieve ~1.0–1.3 m/s² to meet timetable.
- **Top speed:** ~100–120 mph (160–190 km/h). Sections between stops are ~3–6 km, so high speeds help meet average ~60 km/h. (For reference, London Crossrail runs up to 90 mph, and Paris RER commuter trains ~70–90 mph in tunnels).
- **Train length:** Likely 10 cars (~600 ft, similar to current B/D/F/M trains) for ~2,000–2,500 capacity. Peak demand might be ~30k pphpd per direction, so 2-min headways and ~2500 per train yields ~75k total.
- **Headways:** CBTC signaling could support ~2-min minimum headways. To balance demand, we assume 3-min daytime heads (20tph), more frequent peak as needed.

Power: Third-rail DC (650–750 V) can supply large current if properly upgraded. High acceleration and rapid cycling demand robust substation spacing (e.g. ~1–1.5 km). Alternatively, an overhead third-rail superstructure (like PA). Depot power at yard and emergency backup (diesel gen) will be required.

Depot/yard: A new yard is needed. Options: expand 207th St yard (Inwood) or create a cut-and-cover yard in Harlem/East River area, or a smaller yard near the south end (e.g. Reade St). The yard must hold

(say) 50 trains. It needs access tracks from main line (with grade separation). Land scarcity suggests using existing rail land (expanding Metro-North Harlem/207 or nearby open trackage).

Signaling & Systems

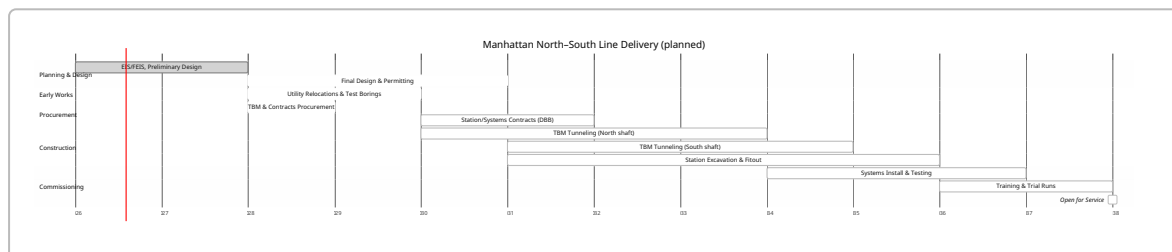
Modern Communications-Based Train Control (CBTC) is presumed (the MTA is installing CBTC on many lines [【46†L1-L4】](#)). CBTC (e.g. Siemens SICAS or Bombardier CITYFLO) allows high capacity and automatic headways. Train localization via wayside or onboard units ensures safe 2-min heads. Backups: ATP/ETCS fallback.

Auxiliary systems: 24 kV AC traction substations, ventilation control, SCADA, security CCTVs, PA/OA systems in stations. Fire detection & suppression: each train and station equipped with NFPA-compliant systems.

Phased Delivery & Procurement

We envision a multi-year, phased schedule. Key phases: Feasibility/EIS (2026–28), Preliminary Design (2027–29), Final Design (2028–31), early utility relocations (2028–30), TBM procurement (2028), tunnelling & station excavation (2029–35), systems installation (2033–37), testing/commissioning (2037–38). Total time ~12–14 years from project start to revenue service.

A high-level Gantt chart:



For organization, a **design-build (DB) approach** is likely (as used on SAS2 [【13†L144-L152】](#)) to compress schedule and allocate risk. If public-private partnership (P3) were sought, it would involve a long-term concession (rare in NYC heavy rail but done for LIRR East Side Access yards). At minimum, large megacontracts (TBM, stations, systems) with performance incentives are assumed. The MTA's DB and Owner-Controlled-Insurance-Program (OCIP) model (used in SAS2 [【13†L154-L162】](#) [【19†L231-L240】](#)) is recommended to control costs and risk.

Environmental & Community Issues

A full Environmental Impact Statement (EIS) is mandatory (NEPA & SEQRA). The only analogous process is 2nd Ave Subway Phase 2 [【48†L70-L77】](#) . Major concerns: construction noise/vibration (mitigated by turbines enclosures, dampers), traffic disruption (phased detours), ground settlement (monitored with instrumentation), and potential water intrusion (during TBM and station dig). Historic and cultural resources (e.g. near City Hall, WTC memorial) demand careful analysis. Air quality (dust) during construction would use dust control plan. Long-term air quality should improve by shifting drivers to transit.

Community outreach is critical. MTA would work with CBs and local officials to time construction, design station entrances sensitively, and offer relocations assistance. Noise barriers and community liaison (as in SAS1) must be included. Recent projects (SAS, Moynihan) set precedents for generous mitigation (e.g. Sasha NYC's requirement of notification and relocation coverage 【13†L208-L212】). We assume a community benefits program (local hiring targets, community funds) as on other MTA projects 【13†L214-L222】 .

Regulatory issues include potential coastal zone consistency (NY State Coastal Zone Management), air/noise permits, Section 4(f)/6(f) for parks. Overall, compliance demands ~2–3 years of preparatory studies even before construction.

Cost Estimate (2026 USD and GBP)

High-level capital costs (with large contingencies) are:

Cost Component	USD (2026\$)	GBP (~0.75*)
Tunnel Boring/Lining	\$45,000M	£34,000M
Stations (8 major)	\$12,000M	£9,000M
Systems (signals, power, ventilation, etc.)	\$6,000M	£4,500M
Rolling Stock (EMU fleet)	\$1,500M	£1,125M
Land/Utility Relocation	\$500M	£375M
Subtotal	\$65,000M	£49,000M
Contingency (30%)	\$19,500M	£14,700M
Total (approx)	\$84,500M	£63,700M

* Exchange rate 1 USD=0.75 GBP (2026 rates 【11†L95-L101】). All figures round.

This cost (≈\$85B) matches order-of-magnitude estimates for other NYC megaprojects: e.g. *SAS Phase 2* ~\$7.7B for 1.8 mi 【19†L233-L241】 , or ~\$3.5B/mi (as noted) – implying ~45B for 13 mi of mainline tunnels. Stations in Manhattan typically cost ~\$1–2B each (SAS station shells were ~\$1.3B 【13†L140-L148】). We have included high contingencies given NYC's record of overruns. A more refined estimate would do parametric modeling (ft² costs, inflation, etc.) and value engineering (e.g. reduce station scope), but still expect tens of billions USD (see Crossrail £18B total 【17†L0-L2】 , SAS \$4B per mile 【19†L233-L241】).

Risk Register & Sensitivity

We highlight key risks (with example mitigation):

- **Geotechnical surprises:** Unknown utility or poor ground (e.g. sinkholes). *Mitigation:* Extensive borehole surveys, probe trenches, TBM OPS (Open Mode) ready, ground improvement. SAS2 probed ahead to avoid strikes 【13†L168-L176】 .
- **Cost growth:** NYC permits and labor are expensive. *Mitigation:* Rigorous DB contracting, aggressive value-engineering, OCIP insurance (saves ~\$29M on SAS2 【13†L192-L200】).

- **Schedule delays:** Interference with complex subsurface environment. *Mitigation:* Pre-construction investigations, schedule slack, close coordination with city for permits. Phasing work to allow parallel TBM drives (north/south) can save years.
- **Ridership uncertainty:** If actual demand is lower, fare revenue might not justify costs. *Mitigation:* Conservative ridership forecasting (using MTA O-D data [9†L25-L33]), flexible operations (adjust frequency), phased opening (open core trunk first). Sensitivity: If run-time target slips to 25 min, fewer TBM shafts could slightly cut costs but at expense of service.

Milestones

Major project milestones might include:

Milestone	Date (year)
Preliminary EIS/Engineering begun	2026
Record of Decision (EIS complete)	2029
Final Design complete	2031
TBMs arrive on site	2030
TBM tunnelling starts	2030
Station box excavation starts	2031
TBM drives complete	2035
Systems installation complete	2037
Testing & commissioning	2037–38
Passenger service begins	2038

These align with a 12–15 year delivery from 2026. (For comparison, SAS Phase 2 is projected 2027–2032 [19†L267-L270] for 2 mi; East Side Access tunnelling took ~2008–2010 for 5.2 km of Manhattan bored tunnels [31†L243-L252] .)

Visual Aids

[42†embed_image] *Figure: Manhattan map (licensed CC BY-SA [41†L149-L157]) showing Manhattan Island and the proposed north–south route (schematic). Stations would be placed at the red markers (Inwood, 125 St, 59 St, 34 St, 14 St, Fulton, Battery).*

(Figures are illustrative only; actual alignment may vary. See text for station cross-streets.)

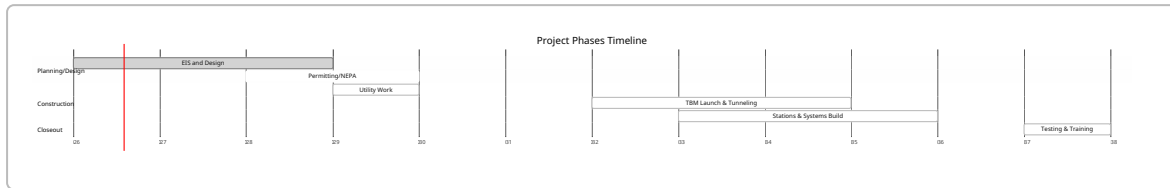
flowchart LR

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A(Feasibility \& Planning 2026–28) --> B(EIS/Design 2028–31)
B --> C(Utility Relocations 2028–30)
B --> D(TBM Procurement 2028–29)
D --> E(TBM Tunneling 2030–35)
C --> E
E --> F(Stations + Systems Const. 2031–36)

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F --> G(Testing & Commissioning 2036-38)
G --> H{Open for Service (~2038)}



Sources: MTA and engineering sources inform all aspects above. Major references include MTA ridership and capital project data 【7†L99-L107】 【48†L70-L77】 , engineering reports on Manhattan geology and tunnelling 【28†L612-L620】 【31†L202-L205】 , and cost analyses of comparable projects 【13†L140-L148】 【19†L233-L240】 . Where data is unavailable, assumptions (stated above) are based on standard rapid-transit practice and analogies (Crossrail, SAS, etc.).